

7.5 Applied Mechanics (Engineering Mechanics)

http://en.wikipedia.org/wiki/Applied_mechanics

Applied Mechanics or Engineering Mechanics as it's more commonly referred to as, is a foundation course for many branches of engineering including Civil and Mechanical. The Wikipedia link will give you a general overview of the subject including interesting historical information.

Statics and Mechanics of Material

<http://claymore.engineer.gvsu.edu/~jackh/eod/egr209.html>

This course is somewhat older (1996-98) but is useful as a quick reference for most of the basic concepts. It also includes an overview of presentation techniques for engineers, which is something that is lacking in most engineering courses. Written and oral presentation of reports is an important part of both academic and professional life. Follow the "Units, Material Properties and a Math Handbook" link to find the instructions on honing your presentation skills. Since this is somewhat old, quite a few of the links are dead or require additional software. The downloadable files with the MCD extension require the MathCAD software.

Self Assessment: www.lboro.ac.uk/faculty/eng/engtlsc/Eng_Mech/tutorials/tut_index.htm

This site allows you to test yourself in several key areas of Statics and Dynamics. Each topic has a set of multiple choice questions which you can answer online and get feedback on your performance with the correct solutions, after you answer the questions. As with all such self assessment systems, it is a good idea to prepare yourself before attempting or even looking at the answers. This will help you get a genuine idea of your grasp of the subject and test your scoring capability.